Python List Slicing Task (20 Questions)

Question 1

Extract the sublist `[1, 2, 3]` from the given list.

Input:

lst = [1, 2, 3, 4, 5, 6]

Expected Output:

[1, 2, 3]

Question 2

Extract the sublist `[4, 5, 6]` from the given list.

Input:

lst = [1, 2, 3, 4, 5, 6]

Expected Output:

[4, 5, 6]

Question 3

Extract elements from index 2 to index 6 (exclusive).

Input:

lst = [10, 20, 30, 40, 50, 60, 70, 80]

Expected Output:

[30, 40, 50, 60]

Question 4

Extract the last 3 elements using a negative start index.

Input:

lst = [10, 20, 30, 40, 50, 60, 70]

Expected Output:

[50, 60, 70]

Question 5

Extract the first 4 elements using only the stop parameter.

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Input:

lst = [100, 200, 300, 400, 500, 600]

Expected Output:

[100, 200, 300, 400]

Question 6

Extract every 2nd element from the list.

Input:

lst = [1,2,3,4,5,6,7,8,9,10]

Expected Output:

[1,3,5,7,9]

Question 7

Extract every 3rd element starting from index 1.

Input:

lst = [10,20,30,40,50,60,70,80,90,100]

Expected Output:

[20,50,80]

Question 8

Reverse the entire list using slicing.

Input:

lst = [1,2,3,4,5]

Expected Output:

[5,4,3,2,1]

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Question 9

Extract elements from index 4 down to index 1 (exclusive of index 1) using a negative step.

Input:

lst = [10,20,30,40,50,60]

Expected Output:

[50,40,30]

Question 10

Extract every 2nd element from index 6 to index 0 (backwards).

Input:

lst = [1,2,3,4,5,6,7,8,9]

Expected Output:

[7,5,3,1]

Question 11

Create a copy of the entire list using slicing.

Input:

lst = [5,10,15,20]

Expected Output:

[5,10,15,20]

Question 12

Extract elements from the beginning up to (but not including) index 7 by omitting the start index.

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Input:

lst = [1,2,3,4,5,6,7,8,9,10]

Expected Output:

[1,2,3,4,5,6,7]

Question 13

Extract elements from index 4 to the end by omitting the stop index.

Input:

lst = [10,20,30,40,50,60,70]

Expected Output:

[50,60,70]

Question 14

What happens when you slice with start and stop indices far outside the list length?

Input:

lst = [1,2,3,4,5]

result = lst[100:200]

Expected Output:

[]

Question 15

What happens when you reverse slice with start > stop but no step specified?

Input:

lst = [1,2,3,4,5]

result = lst[4:1]

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Expected Output:

[]

Question 16

Extract every 3rd element starting from index 4 up to (but not including) index 20.

Input:

lst = [10,20,30,40,50,60,70,80,90,100,110,120]

Expected Output:

[50,80,110]

Question 17

Extract the last 5 elements, then reverse them using a single slice.

Input:

lst = [1,2,3,4,5,6,7,8,9,10]

Expected Output:

[10,9,8,7,6]

Question 18

Extract all elements except the first and last using slicing.

Input:

lst = [100,200,300,400,500,600]

Expected Output:

[200,300,400,500]

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Question 19

Extract every element with an even index (0,2,4...) using slicing.

Input:

lst = [10,20,30,40,50,60,70,80]

Expected Output:

[10,30,50,70]

Question 20

Extract elements from index 3 to index 9 with a step of 2, then reverse the extracted result.

Input:

lst = [10,20,30,40,50,60,70,80,90,100]

Step 1

Extract from index **3** to **9** with **step = 2**

Step 2

Reverse the extracted list.

Expected Output:

[80,60,40]

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